Tab 1

N. Shara, R. Mirabal-Beltran, B. Talmadge, N. Falah, M. Ahmad, R. Dempers, S. Crovatt, S. Eisenberg, and K. Anderson, “Use of Machine Learning for Early Detection of Maternal Cardiovascular Conditions: Retrospective Study Using Electronic Health Record Data”, Vol.8, April 2024.

HOPE-CAT trained on blood pressure, heart rate, symptoms, race, ethnicity, age, medical history, family history, and other physical findings. The patient data was sampled from a population of patients with multiple pregnancy-related medical encounters from the ages 18 to 40. Risk profiles were created as outputs of the model that linked risk factors to syndromes. From 604 patients with diagnoses that are able to be compared with, HOPE-CAT was able to detect preeclampsia risk factors 60.2 days before diagnosis.

L. Butler, F. Gunturkun, L. Chinthala, I. Karabayir, M. S. Tootooni, B. Bakir-Batu, T. Celik, O. Akbilgic, and R. L. Davis, “AI-based Preeclampsia Detection and Prediction with Electrocardiogram Data”, March 2024.

Prolonged QT interval, altered p-wave duration, and LV strain are common among preeclampsia patients. ResNet CNN used to predict preeclampsia based on 12-lead 10 second ECGs (AUC = 0.81, Accuracy = 0.78, Precision = 0.65, Sensitivity = 0.83, Specificity = 0.66) on holdout testset. The AUC was 0.79 30+ days before preeclampsia diagnosis.

A. Bulez, K. Hansu, E.S. Cagan, A.R. Sahin, and H.O. Dokumaci, “Artificial Intelligence in Early Diagnosis of Preeclampsia”, March 2024.

LightGBM is used as the model and runs on data of age, blood tests, blood pressure, and various physical statuses and conditions. Main features that were important were results from the blood test and age (AUC = 0.832, Accuracy = 0.906, Sensitivity = 0.737, Specificity = 0.927).

Y. Zhang, K. G. Sylvester, B. Jin, R. J. Wong, J. Schilling, C. J. Chou, Z. Han, R. Y. Luo, L. Tian, S. Ladella, L. Mo, I. Maric, Y. J. Blumenfeld, G. L. Darmstadt, G. M. Shaw, D. K. Stevenson, J. C. Whitin, H. J. Cohen, D. B. McElhinney, and X. B. Ling, “Development of a Urine Metabolomics Biomarker-Based Prediction Model for Preeclampsia during Early Pregnancy”, May 2023.

XGBoost is used as the model and runs on data collected from urine samples which after performing feature analysis, 7 compounds were used (AUC = 0.856, Sensitivity = 0.765, before normalization). Data was a small population of 60 pregnant women.

J. M. Shantanu, H. R. Shaik, S. R. Sharma, P. Strong, U. Srivatsa, I. Ebony, H. Jeong, C-N. Chuah, and L. Mo, “Linking Electrocardiogram and Echocardiogram: Comparing Classical Machine Learning and Deep Learning Neural Networks for the Detection of Regional Wall Motion Abnormalities”, July 2025.

12-lead 10-second ECG Nightingale Dataset with RWMA as ground truth. Inference occurs on an undersampled balanced dataset using three models, one expert featurization using Random Forest and one that is a 1D CNN. Main features used were QT duration and maximum amplitude differential for RF (AUROC = .6358, AUPRC = .7264, Accuracy = .6358, F1-Score = .6319, Precision = .6389, Recall = .6276). 1D CNN was 11 layers and performed best for detecting RWMA (AUROC = 0.734, AUPRC = 0.7245, Accuracy = 0.7115, F1-Score = 0.6956, Precision = 0.7508, Recall = 0.6517). Shows that ECG data can be used to detect an abnormality.

J. Xue, T. Dwivedi, K. Barnett, A. Vlasenko, M. Kirsch, A. Parekh, R. Yuan, G. Clifford, and D. Albert, “Does a Reduced ECG Lead Set Contain the Full 12-lead ECG Information for Interpretation?”, Computing in Cardiology, Vol. 51, 2024.

Leads I and II are used to generate leads III, aVR, aVL, and aVF and passed through more layers to output morphology interpretations from the ECG.The reduced lead set with only leads I, II, V2 and V4 perform close to the full 12-lead set. The synthetic additions of V1, V3, V5, and V6 did not produce any improvement. Reduced ECG is enough to detect morphological abnormalities.

F. Musa and R. Prasad, “Predicting Preeclampsia Using Principal Component Analysis and Decision Tree Classifier”, Current Women’s Health Review, 2024.

XGBoost is used on data from EHR, blood pressure, and urine samples (Evaluation metrics were extremely good across different samplings of the dataset).
Problem with the dataset: no idea when the diagnosis or data is collected like if they were collected before or after preeclampsia was diagnosed.

Tab 2

Related/Previous Work/Literature Review

Decision tree-based models have proven to be successful in predicting preeclampsia, though none through the use of ECGs for inference. Bulez et al observed the use of a LightGBM model using data from blood tests, blood pressure, and other factors from patient EHRs with the conclusion that age and features from the blood test were most important for inference [1]. In Zhang et al's work, the XGBoost model was used on urine samples to determine seven compounds that were most important in predicting preeclampsia [2].

Evaluation Criteria

As a binary classification problem, the main metrics used in determining the model performance will be area under curve (AUC), accuracy, precision, recall, and F1-score. MUSE values of certain features are reported from the UCDH Echo dataset which will be compared to features created locally from Neurokit to determine accurate implementation. Once the featurizations have been implemented, feature importance analysis will be used to determine which features from the ECGs are most useful for prediction, ultimately pruning the features by analyzing tradeoffs from the performance metrics.

Implementation and Results

Without current access to the UCDH Echo dataset, implementing the ECG features for the RF model is what is being done. The Python library, Neurokit has methods for cleaning ECG signals along with extracting certain features. Through the use of these methods, we are able to implement the features in Table _. Hana Shaik, our graduate mentor, has already implemented ST-T abnormalities and used these features in detecting RWMA. Testing occurred on the Nightingale ECG dataset to ensure that featurization was occurring. As another form of validation, we were able to compare our implementation to Hana's implementation. (Want to include more after 4/14 meeting such as discussion of QRS-axis, T-axis)

Features	Equations
Pmax	max(max(P offsets - P onsets) for 12 leads)
Pmin	min(min(P offsets - P onsets) for 12 leads)
P-Wave Dispersion	Pmax - Pmin
QT Dispersion	QTmax = max(max(T offsets - R onsets) for 12 leads) QTmin = min(min(T offsets - R onsets) for 12 leads) Qtmax - Qtmin

Tp-e Interval	avg(T offsets - T peaks)
Tp-e/QT	(Tp-e Interval) / (avg(T offsets - R onsets)
Tp-e/QTc	Tp-e/QT * sqrt(diff(R Peaks))
ST Elevation	1 if > 50% elevated 0 otherwise
ST Depression	1 if > 50% depressed 0 otherwise
Avg. Amp. Diff.	avg(ST Amp. - PR Amp.)
Med. Amp. Diff.	median(ST Amp. - PR Amp.)
Max. Amp. Diff.	max(ST Amp. - PR Amp.)
Min. Amp. Diff.	min(ST Amp. - PR Amp.)
QRS Axis	TBD
T Axis	TBD
Frontal QRS-T Angle	abs(QRS Axis - T Axis)

Future Work

Once the UCDH Echo dataset and MUSE values are released, we will be able to compare Neurokit values from MUSE values and determine if there are any drastic differences. An RF model will be initially used, but other decision tree models will be tested such as LightGBM and XGBoost to achieve best performance. After model selection, feature importance and feature distribution will be analyzed through χ^2 testing or principal component analysis (PCA). Because the dataset is imbalanced, we may employ SMOTE to train on a balanced dataset.

Acknowledgements

References

[1] A. Bulez, K. Hansu, E.S. Cagan, A.R. Sahin, and H.O. Dokumaci, “Artificial Intelligence in Early Diagnosis of Preeclampsia”, March 2024.
[2] Y. Zhang, K. G. Sylvester, B. Jin, R. J. Wong, J. Schilling, C. J. Chou, Z. Han, R. Y. Luo, L. Tian, S. Ladella, L. Mo, I. Maric, Y. J. Blumenfeld, G. L. Darmstadt, G. M. Shaw, D. K. Stevenson, J. C. Whitin, H. J. Cohen, D. B. McElhinney, and X. B. Ling, “Development of a Urine Metabolomics Biomarker-Based Prediction Model for Preeclampsia during Early Pregnancy”, May 2023.